

Bone Marrow Failure — The Empty Factory



1. BMF entrance hall

WHAT YOU SEE

Central archway labeled BMF — Central Entrance Hall

WHAT IT ENCODES

Bone marrow failure is the unifying concept: the marrow factory fails to produce adequate mature blood cells, manifesting as one or more cytopenias. The central diagnostic question is WHY production is failing — empty marrow (aplastic anemia), dysplastic marrow (MDS), replaced marrow (myelophthisis), or selective failure of one lineage (PRCA). The smear plus the marrow biopsy partition these.

BOARD TRAP

Failure of PRODUCTION (low reticulocytes, hypoproliferative) must be separated from peripheral DESTRUCTION/sequestration (high reticulocytes, hemolysis, hypersplenism) — checking the reticulocyte count is the first branch point and a perennial exam discriminator.

2. Empty AA conveyor

WHAT YOU SEE

Yellow fat-covered idle conveyor belt under the AA sign with question mark

WHAT IT ENCODES

Aplastic anemia is the prototype production failure: an empty, fat-replaced (hypocellular) marrow producing pancytopenia. Most cases are acquired and immune-mediated — cytotoxic T cells attack hematopoietic stem cells (often triggered by drugs, viruses like hepatitis/EBV, or idiopathic). Treatment is immunosuppression (ATG + cyclosporine) for older/no-donor patients, or allogeneic HSCT for young patients with a matched sibling donor.

BOARD TRAP

Because acquired AA is immune-mediated, it RESPONDS to immunosuppression — distinguishing it from inherited (Fanconi/DC) and from MDS matters, since the latter need different therapy and AA-like immunosuppression can be harmful.

3. Pancytopenia bins

WHAT YOU SEE

Three colored bins labeled R, W, P

WHAT IT ENCODES

Pancytopenia = simultaneous reduction of all three lineages: Red cells (anemia), White cells (leukopenia/neutropenia), and Platelets (thrombocytopenia). The differential spans decreased production (AA, MDS, myelophthisis, megaloblastic anemia, marrow infiltration) and increased destruction/sequestration (hypersplenism, PNH, autoimmune).

BOARD TRAP

Pancytopenia is NOT a single disease but a finding — don't equate it with aplastic anemia. An isolated single-lineage cytopenia (e.g., PRCA = red only) points to a completely different, more targeted differential.

4. Hypocellular marrow

WHAT YOU SEE

HYPOCELLULAR sign with empty bin and a yeast/fat clump

WHAT IT ENCODES

The biopsy hallmark of aplastic anemia is a markedly hypocellular marrow (<25% cellularity, or <30% with severity criteria) in which fat replaces hematopoietic tissue, WITHOUT dysplasia, fibrosis, or infiltrate. Severity (Camitta criteria) is graded by neutrophil, platelet, and reticulocyte counts.

BOARD TRAP

A hypocellular marrow is the OPPOSITE of MDS and myelophthisis, which are typically HYPERcellular (MDS) or packed/fibrotic (myelophthisis) — cellularity on biopsy is a key fork. Beware hypocellular MDS, a mimic that needs cytogenetics to unmask.

5. AA exclusion sign

WHAT YOU SEE

AA placard with no-entry / crossed icons and businessman

WHAT IT ENCODES

Aplastic anemia is a diagnosis of EXCLUSION. Before labeling it, rule out hypocellular MDS (cytogenetics, dysplasia), PNH (flow cytometry for CD55/CD59 loss), hairy cell leukemia (causes dry tap + monocytopenia), inherited syndromes (chromosome breakage, telomere length), and reversible causes (drugs, viral, nutritional).

BOARD TRAP

Skipping the exclusion workup — especially PNH flow cytometry and cytogenetics — is the classic error; a small PNH clone or an MDS clone in 'AA' changes prognosis and therapy.

6. Magnifying question

WHAT YOU SEE

Large magnifying glass over a red question mark

WHAT IT ENCODES

The diagnostic approach interrogates marrow architecture: is it EMPTY (aplastic), DYSPLASTIC (MDS), INFILTRATED/fibrotic (myelophthisis), or selectively absent in one lineage (PRCA)? CBC + smear + reticulocyte count narrow it, and the marrow aspirate/biopsy with cytogenetics and flow cytometry confirms.

7. MDS dysplastic line

WHAT YOU SEE

MDS sign over conveyor with abnormal purple grape-like cells

WHAT IT ENCODES

Myelodysplastic syndromes are clonal stem-cell disorders of older adults featuring dysplastic, INEFFECTIVE hematopoiesis — peripheral cytopenias despite a typically HYPERcellular marrow (cells die before release). Findings include dysplastic morphology, ring sideroblasts, and clonal cytogenetic abnormalities (e.g., del(5q), -7). Carries a risk of transformation to AML; prognosis is stratified by IPSS-R.

BOARD TRAP

MDS is the great mimic of aplastic anemia but is typically HYPERcellular with dysplasia and clonal cytogenetics — missing it (and treating as AA) delays appropriate therapy (hypomethylating agents, lenalidomide for del(5q), transplant).

8. Chromosome panels

WHAT YOU SEE

Row of framed XX/XY karyotype panels on the wall

WHAT IT ENCODES

Cytogenetics and clonal markers are decisive in classifying marrow failure: an abnormal clonal karyotype (del(5q), monosomy 7, trisomy 8) signals MDS rather than idiopathic aplastic anemia, refines MDS prognosis, and helps detect clonal evolution. Flow cytometry for a PNH clone complements karyotyping.

BOARD TRAP

A normal karyotype does not exclude MDS (single-lineage dysplasia or molecular mutations may still be present), and finding a clone in apparent AA reclassifies the disease — always send cytogenetics on the marrow.

9. PRCA red-only belt

WHAT YOU SEE

PRCA gateway over a belt carrying only red discs, padlocked

WHAT IT ENCODES

Pure red cell aplasia is selective failure of ONLY the erythroid line: severe anemia with reticulocytopenia but NORMAL white cells and platelets, and a marrow showing absent erythroid precursors with intact myeloid/megakaryocytic lineages. Causes include parvovirus B19, thymoma, drugs, autoimmune/lymphoproliferative disease (CLL, LGL leukemia), and anti-EPO antibodies.

BOARD TRAP

PRCA is single-lineage (red only) — calling it 'pancytopenia' is wrong. Its isolated reticulocytopenic anemia with normal WBC/platelets is the discriminator from aplastic anemia and MDS.

10. Cellular catalog wall

WHAT YOU SEE

CELLULAR sign with full grid of colored cell tiles

WHAT IT ENCODES

A normo- or hypercellular marrow in a patient with cytopenias points AWAY from aplastic anemia toward ineffective hematopoiesis (MDS, megaloblastic anemia), peripheral destruction/sequestration, or infiltration. Marrow cellularity is therefore a central triage feature.

BOARD TRAP

Cytopenias + cellular marrow ≠ aplastic anemia — think MDS, B12/folate deficiency (megaloblastic), or hypersplenism instead. Reserve 'aplastic' for the genuinely hypocellular, fat-replaced marrow.

11. Production failure poster

WHAT YOU SEE

Red PRODUCTION FAILURE and DESTRUCTION posters with workers fighting

WHAT IT ENCODES

The first conceptual split in any cytopenia is reduced PRODUCTION versus increased peripheral DESTRUCTION/sequestration. Low reticulocyte count indicates a hypoproliferative/production problem (the BMF disorders); a high reticulocyte count points to hemolysis or blood loss; splenomegaly suggests sequestration.

BOARD TRAP

BMF is a production (output) problem — don't attribute pancytopenia to hemolysis or splenic sequestration without checking the reticulocyte count and spleen size; the reticulocyte index is the cheap, decisive first test.

12. Crab-shaped shadow

WHAT YOU SEE

DNA helix and crab-shaped shadow near collapsing belt

WHAT IT ENCODES

Behind acquired marrow failure lurk inherited/constitutional syndromes (DNA helix — Fanconi anemia, telomere disorders) and the risk of clonal evolution to MDS/AML (the crab = malignancy). Apparent 'idiopathic' AA can harbor a germline syndrome or evolve clonally over time, mandating ongoing surveillance.

BOARD TRAP

Even acquired aplastic anemia carries a real risk of clonal evolution to MDS/AML and of harboring PNH clones — long-term monitoring is required, and a young AA patient still needs inherited-syndrome screening before transplant.

13. Specialist desk

WHAT YOU SEE

SPECIALIST ONLY desk with clinician and microscope

WHAT IT ENCODES

Definitive classification of bone marrow failure requires a hematologist-performed marrow ASPIRATE plus core BIOPSY, with morphology, cellularity assessment, cytogenetics, flow cytometry (PNH clone), and — when indicated — germline/molecular testing. This is what separates AA from MDS, myelophthisis, and PRCA.

BOARD TRAP

Peripheral blood alone cannot classify marrow failure — the aspirate may give a dry tap (fibrosis/infiltration), so a core biopsy is mandatory; never finalize the diagnosis on CBC and smear alone.